

NPTEL Course on "Computer System Optimization for ML"

Lecture #55

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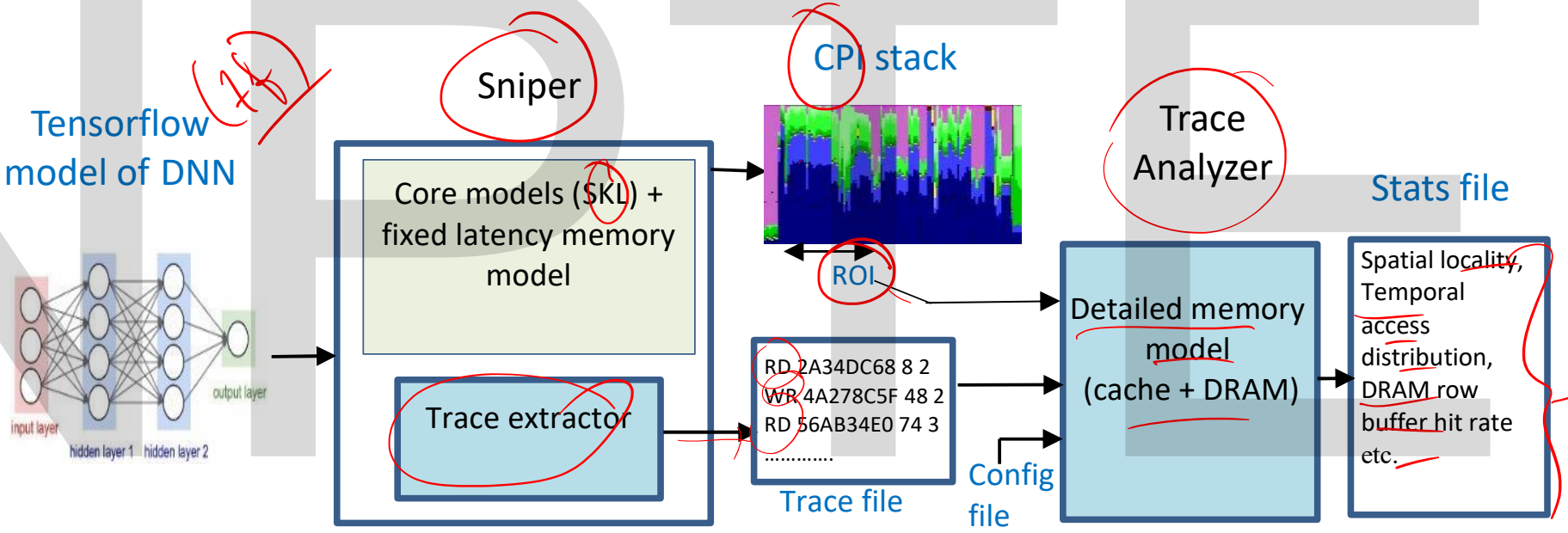
Agenda

- **Performance Characterization of Deep Learning Workloads (contd.)**

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Memory Characterization Framework [1]

- **To capture the complex feedback loop between the CPU and the memory**
- **Systematic framework for memory system characterization of DNN workloads**

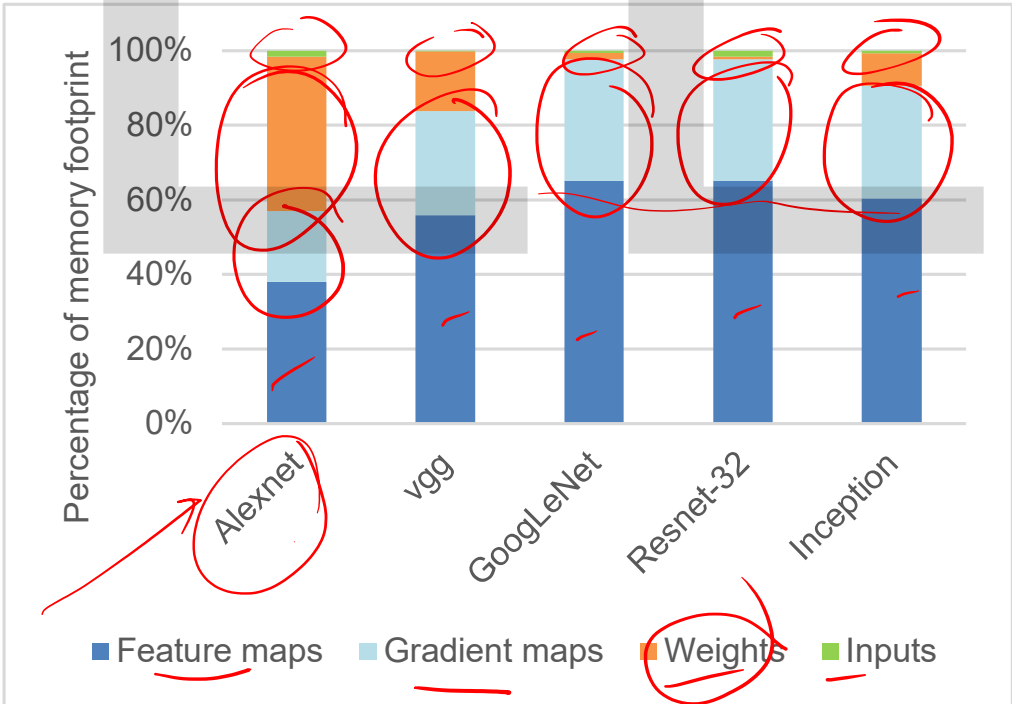


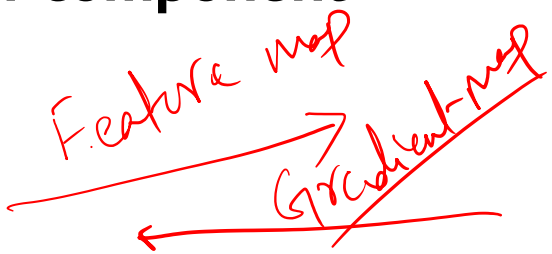
[1] Chishti, Zeshan, and Berkin Akin. "Memory system characterization of deep learning workloads." *Proceedings of the International Symposium on Memory Systems*. 2019.

Memory Requirements

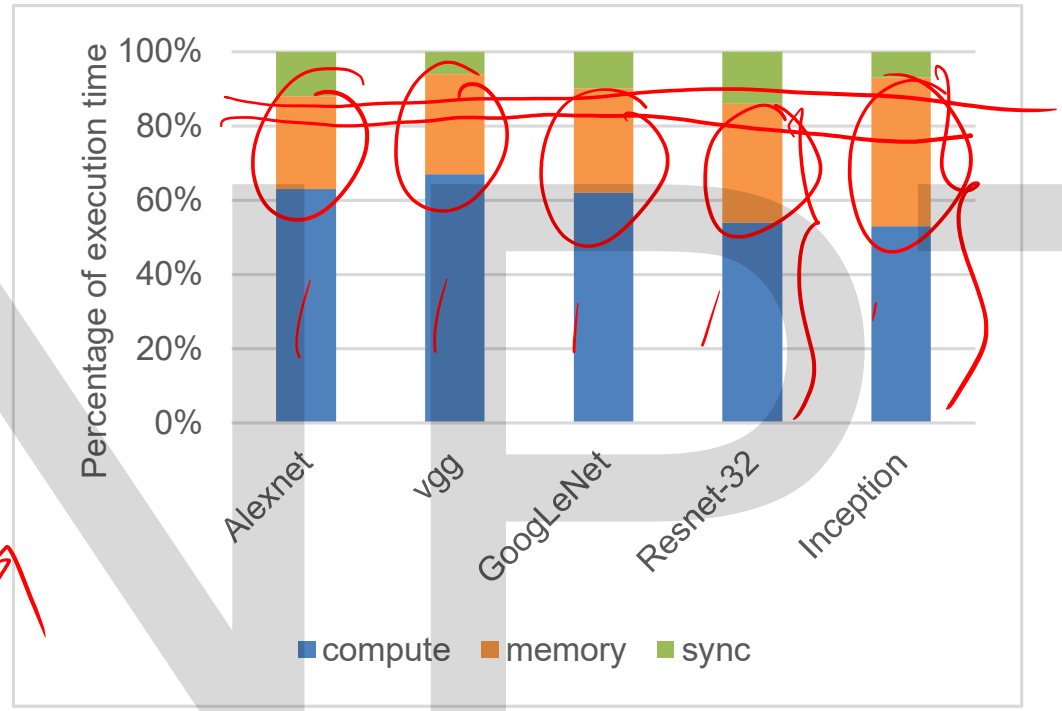
- **Memory capacity and bandwidth requirements of a DNN workload arise from**
  - Inputs
  - Weights
  - Feature maps
- **Cross-layer feature map data accounts for majority of the memory footprint**
- **Gradient maps communication during backward propagation is another major component**

Workload	No. of layers	Batch size	Memory footprint (GB)	Per-core IPC	LLC MPKI
Alexnet	8	64	0.6	1.63	2.6
vgg	19	32	3.5	1.77	7.7
GoogLeNet	22	64	1.5	1.35	4.8
Resnet-32	34	128	0.25	1.33	2.9
Inception-resnet-v2	22	32	1.2	1.82	7.0

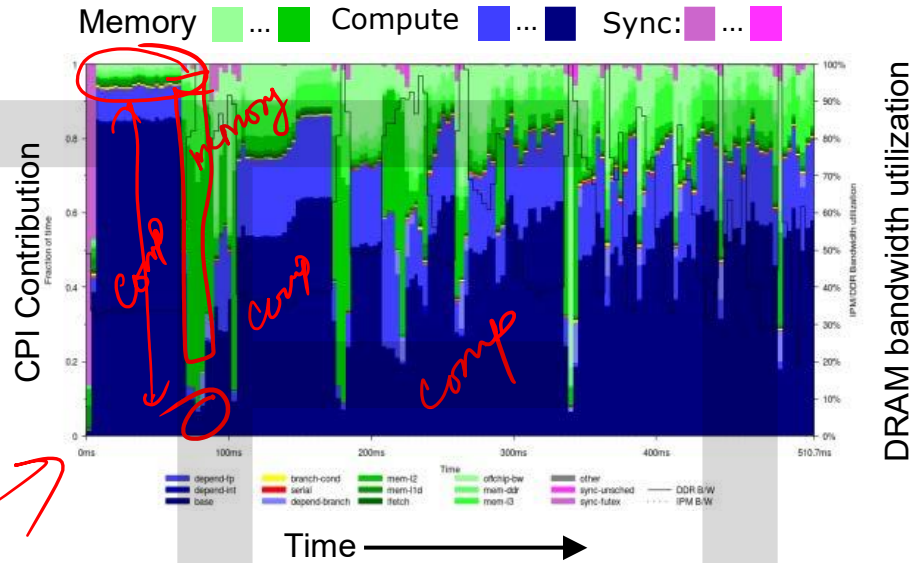




Compute vs. Memory Intensity



Memory accesses constitute a significant portion of execution time – 25% to 40% across workloads

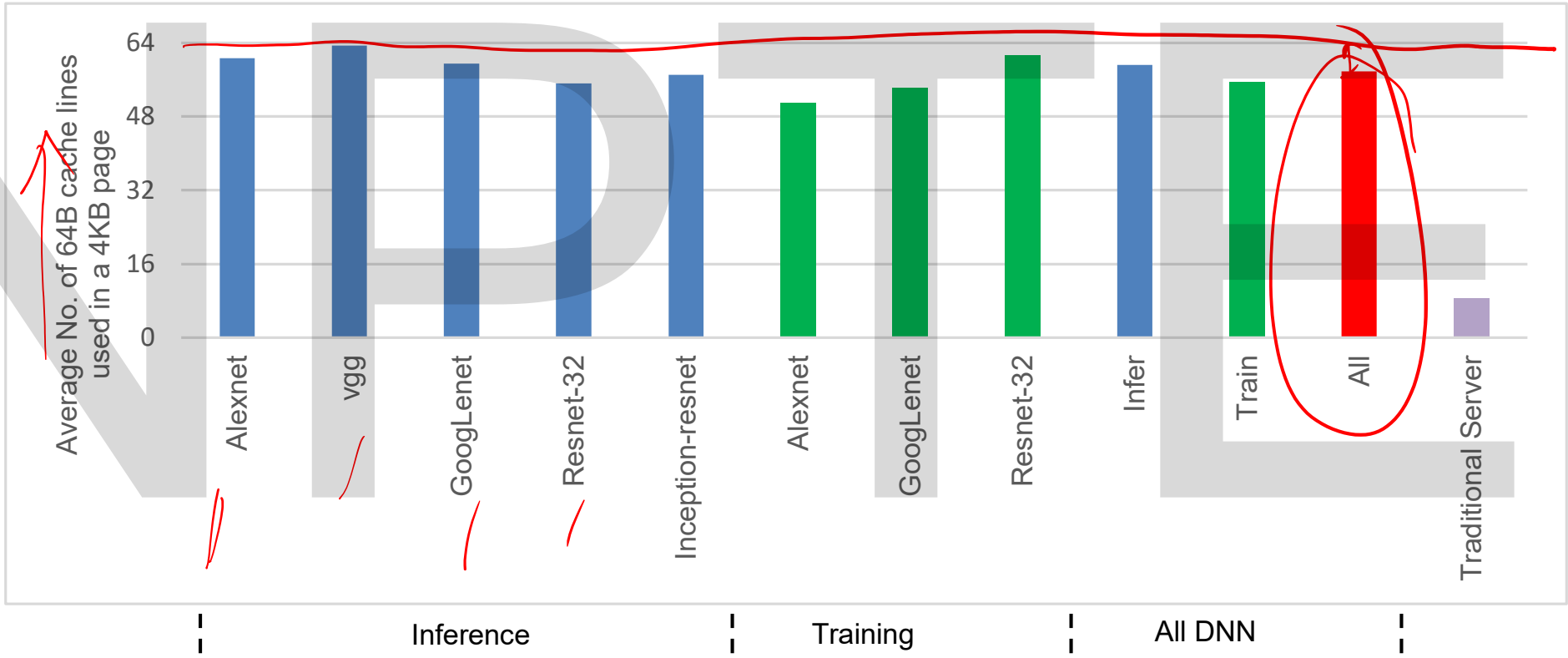


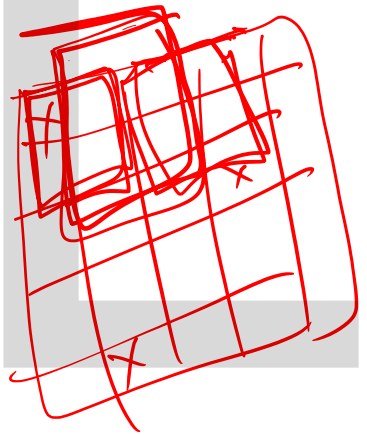
Distinct compute-intensive ("blue" dominated) and memory-intensive ("green" dominated) phases

NDT

Spatial Locality

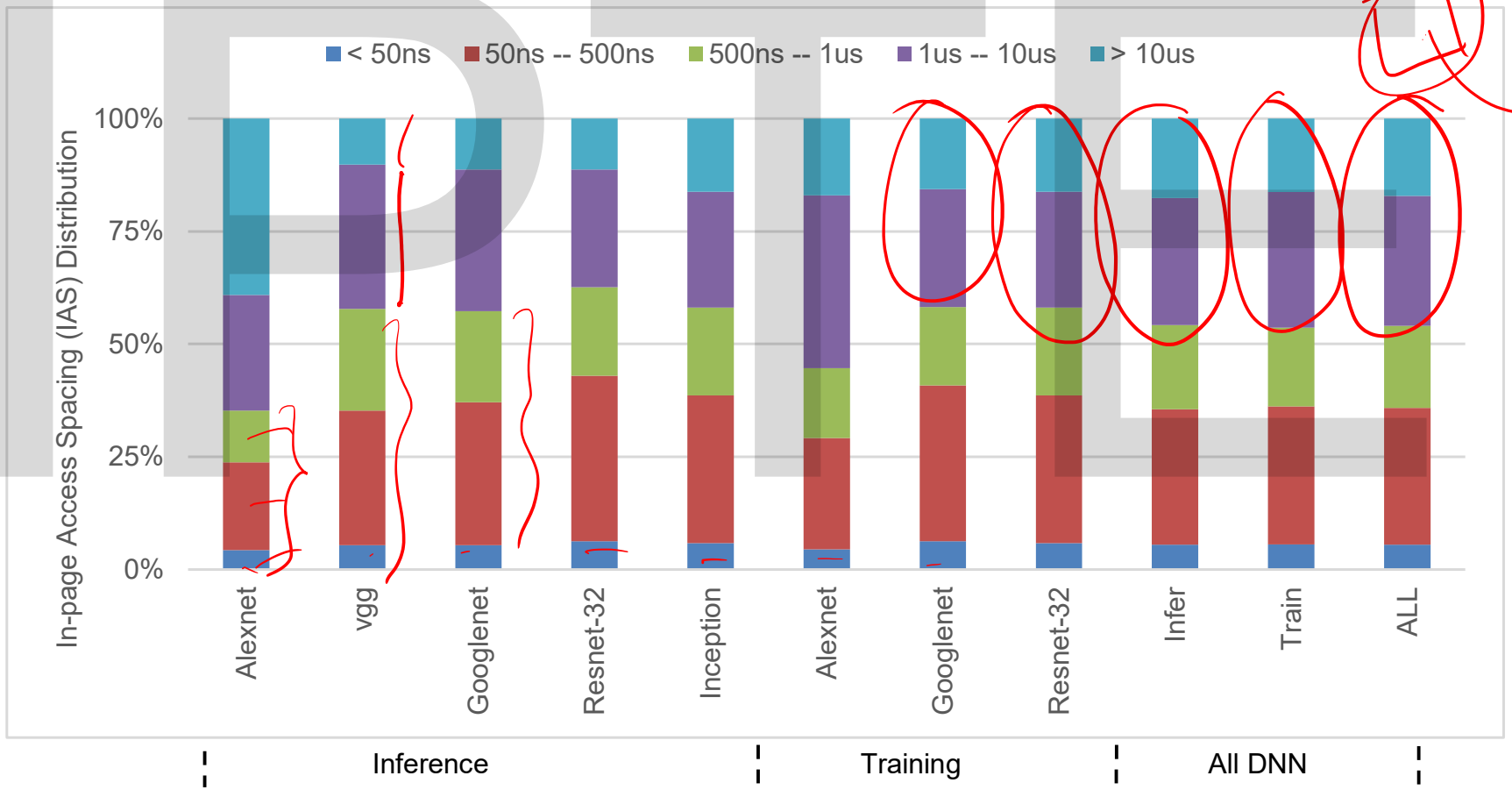
- **Significant number of cache lines in 4KB page being accessed**





Spatio-Temporal Locality

- **Inter-Line Access (In page Access Spacing)**
- **~50% accesses are more than 1uS apart!**

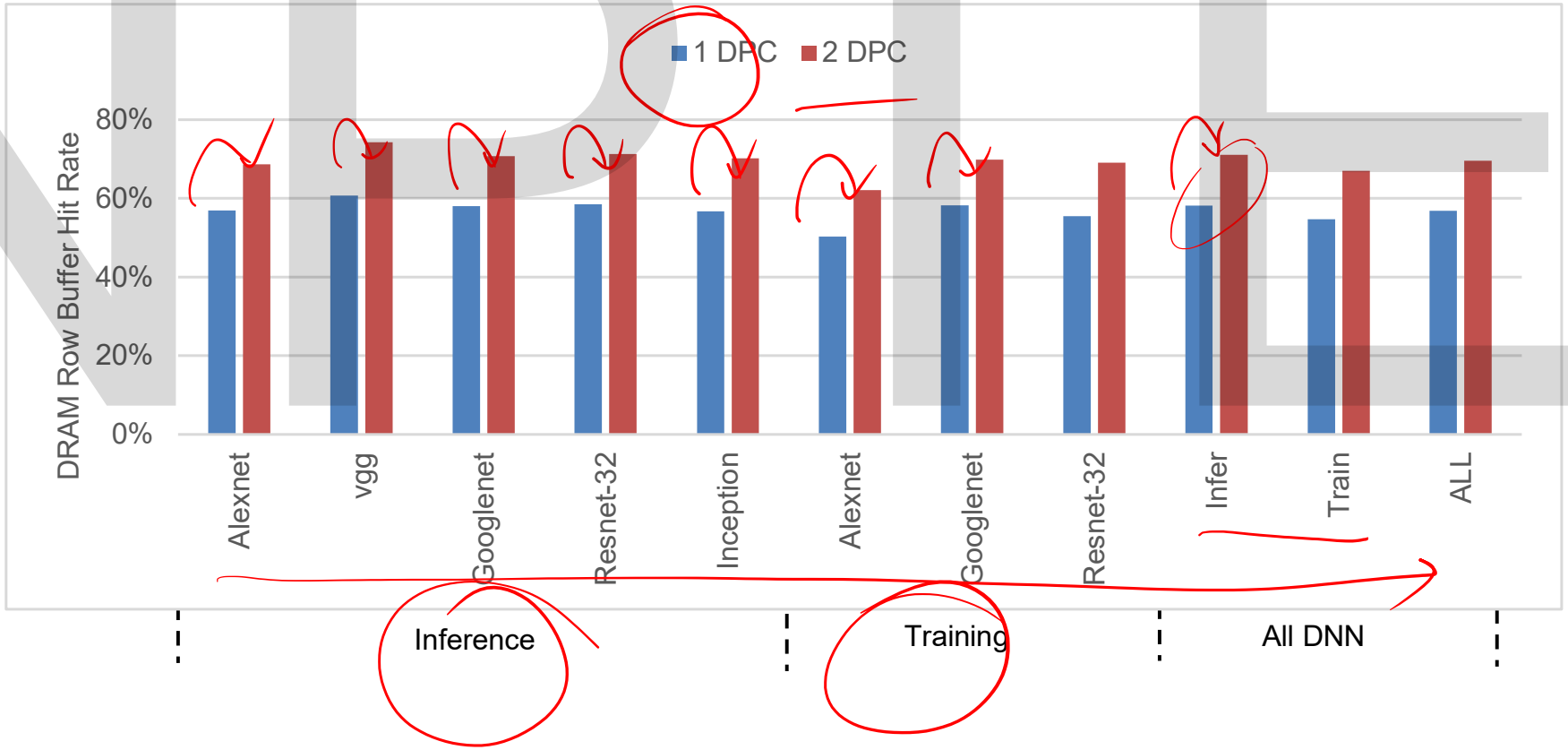


NDTL

Row Buffer Hit Rates

- **High RBH → Mostly streaming data!**
- **More DRAM → Higher RBH**

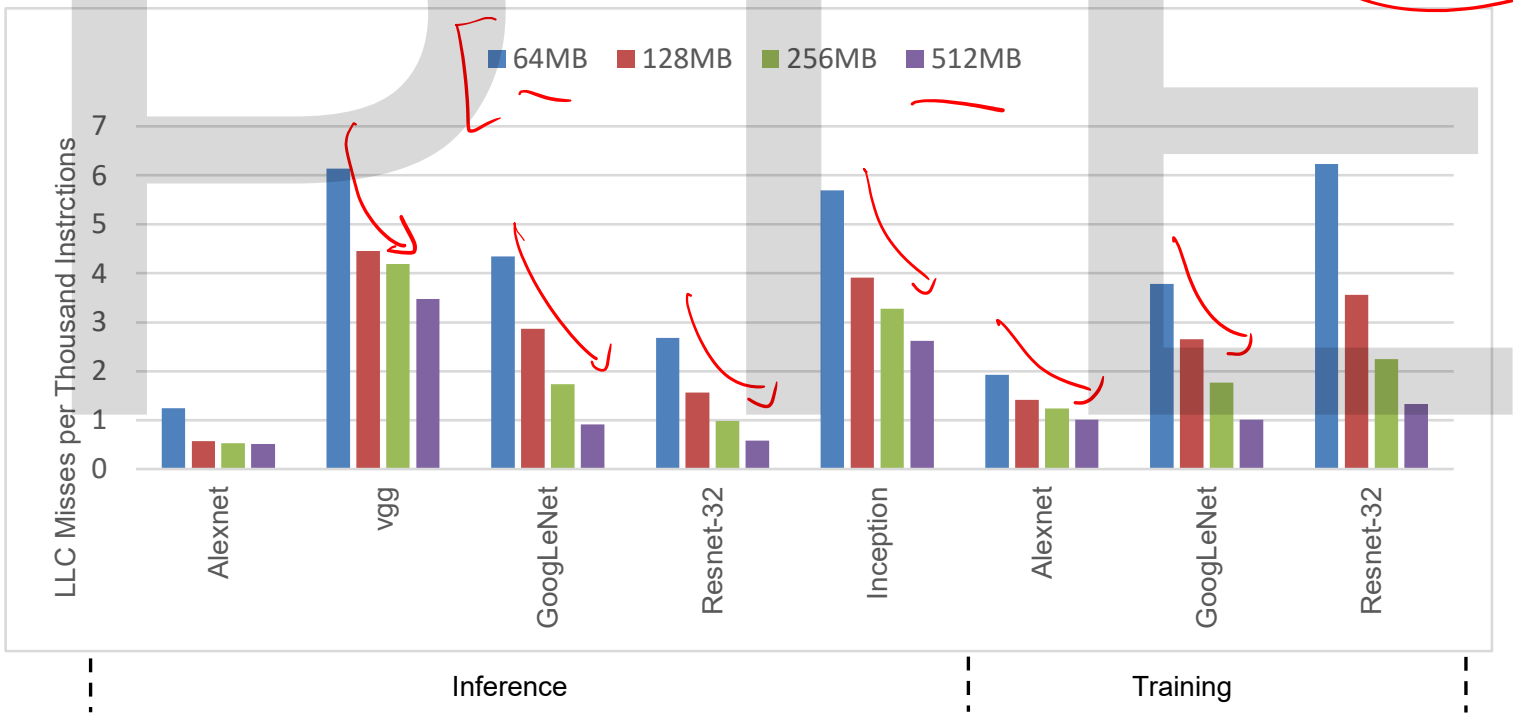
- **Two configurations**
  - 1 DDR4 DIMM per channel
  - 2 DDR DIMMs per channel



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LLC Sensitivity

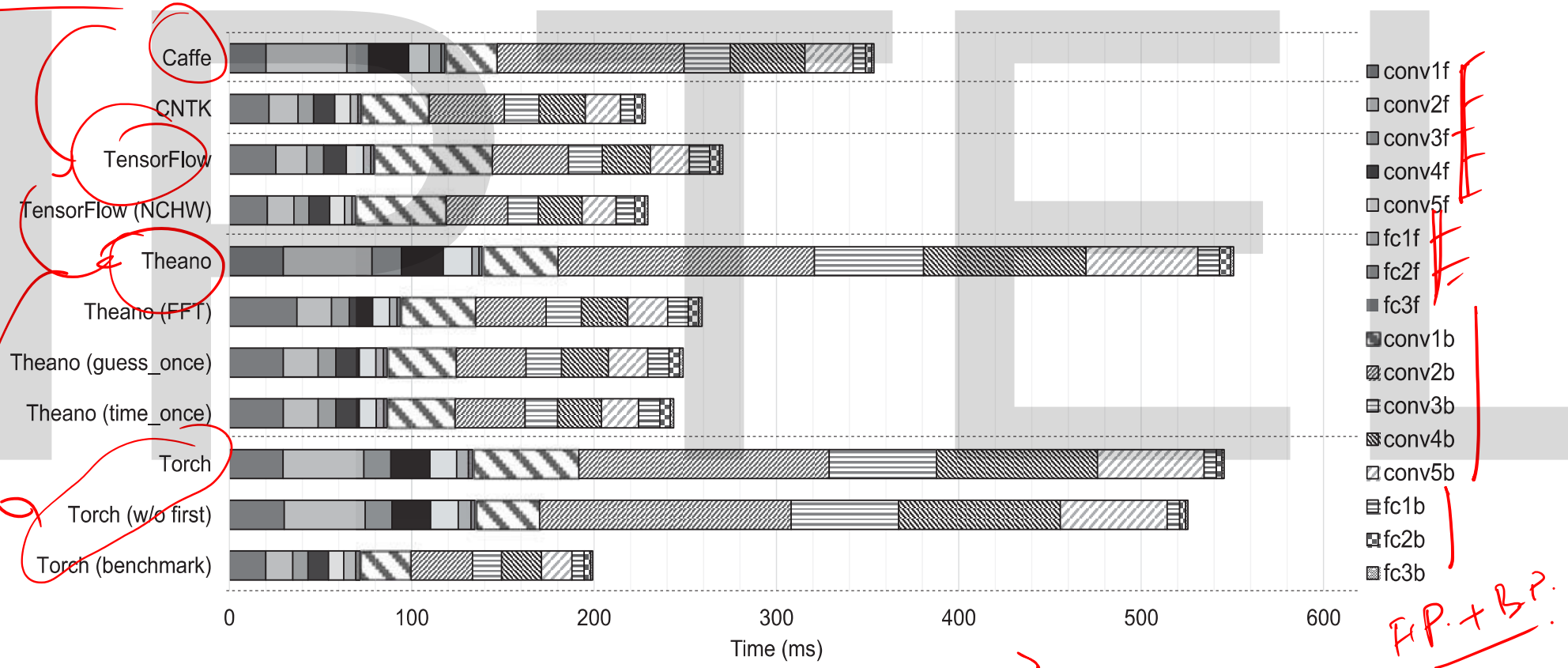
- Larger LLCs can fit intermediate data (feature maps), avoiding costly DRAM accesses
  - Using eDRAM or other dense memory technologies (such as STTRAM) to build LLCs



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Further Studies [2]

- **Performance of DNN (AlexNet) under different frameworks: Pytorch vs.Caffe vs. Tensorflow vs. …**



[2] Kim, Heehoon, et al. "Performance analysis of CNN frameworks for GPUs." *2017 IEEE international symposium on performance analysis of systems and software (ISPASS)*. IEEE, 2017.

Further Studies [3]

- **Inference time characterization on different GPUs**
  - **Titan Xp and Jetson TX1**

Batch Size

DNN	1	2	4	8	16	32	64
AlexNet	1.28	0.70	0.48	0.27	0.18	0.14	0.15
BN-Inception	3.79	3.00	1.64	1.10	0.87	0.77	0.71
CaffeResNet-101	8.20	4.82	3.32	2.54	2.27	2.16	2.08
DenseNet-121 (k=32)	8.93	4.41	2.64	1.96	1.64	1.44	1.39
DenseNet-169 (k=32)	13.03	6.72	3.97	2.73	2.14	1.87	1.75
DenseNet-201 (k=32)	17.15	9.25	5.36	3.66	2.84	2.41	2.27
DenseNet-161 (k=48)	15.50	9.10	5.89	4.45	3.66	3.43	3.24
DPN-68	10.68	5.36	3.24	2.47	1.80	1.59	1.52
DPN-98	22.31	13.84	8.97	6.77	5.59	4.96	4.72
DPN-131	29.70	18.29	11.96	9.12	7.57	6.72	6.37
FBResNet-152	14.55	7.79	5.15	4.31	3.96	3.76	3.65
GoogLeNet	4.54	2.44	1.65	1.06	0.86	0.76	0.72
Inception-ResNet-v2	25.94	14.36	8.82	6.43	5.19	4.88	4.59
Inception-v3	10.10	5.70	3.65	2.54	2.05	1.89	1.80
Inception-v4	18.96	10.61	6.53	4.85	4.10	3.77	3.61
MobileNet-v1	2.45	0.89	0.68	0.60	0.55	0.53	0.53
MobileNet-v2	3.34	1.63	0.95	0.78	0.72	0.63	0.61

Titan Xp

DNN	1	2	4	8	16	32	64
AlexNet	28.88	13.00	8.58	6.56	5.39	4.77	
BN-Inception	35.52	26.48	25.10	23.89	21.21	20.47	
CaffeResNet-101	84.47	91.37	70.33	63.53	56.38	53.73	
DenseNet-121 (k=32)	66.43	50.87	50.20	43.89	40.41	38.22	
DenseNet-169 (k=32)	137.96	130.27	110.82	100.56	92.97	88.94	
DenseNet-201 (k=32)	84.57	61.71	62.62	53.73	49.28	46.26	
DenseNet-161 (k=48)	103.20	76.11	77.10	68.32	62.73	59.14	
DPN-68	113.08	52.73	42.85	43.32	38.18	36.40	36.22
DPN-98	243.51	148.51	135.30	125.92	123.34	118.68	117.27
DPN-131	330.15	204.69	184.89	172.25	165.59	162.67	160.66
FBResNet-152	133.68	147.75	113.48	105.78	94.26	97.47	
GoogLeNet	32.11	27.19	23.29	21.66	19.77	19.96	
Inception-ResNet-v2	198.95	141.29	127.97	130.25	117.99	116.47	
Inception-v3	79.39	59.04	56.46	51.79	47.60	46.85	
Inception-v4	158.00	120.23	106.77	102.21	95.51	95.40	
MobileNet-v1	15.06	11.94	11.34	11.03	10.82	10.58	10.55
MobileNet-v2	20.51	14.58	13.67	13.56	13.18	13.10	12.72

Jetson TX1

[3] Bianco, Simone, et al. "Benchmark analysis of representative deep neural network architectures." *IEEE access* 6 (2018): 64270-64277.



THANK YOU

Slide Courtesy: Murali Annavaram (University of Southern California), R. Govindarajan (IISc), Arkaprava Basu (IISc), Jayashree Mohan (Microsoft Research)